

Selective Guidance Network with edge and texture awareness for polyp segmentation

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ABSTRACT

Colorectal polyp segmentation plays a crucial role in preventing colorectal cancer through colonoscopic image screening. However, most existing methods overlook uncertain regions in colonoscopic images, particularly the blurred boundary areas where polyps closely resemble colon fold structures. To address this challenge, we propose the Selective Guidance Network with edge and texture awareness for polyp segmentation (SGNet). SGNet consists of three essential modules, namely the Edge and Texture Awareness Module (ETAM), the Prior Enhancement Module (PEM), and the Hierarchical Feature Fusion Module (HFFM). ETAM integrates Laplacian operators with spatial attention mechanisms to enhance feature perception, allowing for precise extraction of polyp boundaries and adaptive amplification of texture patterns. PEM strengthens multi-scale contextual perception through dilated convolutions while refining backbone features through dual prior-driven feature rectification. HFFM employs multi-level attention gating to achieve cross-scale feature integration while effectively combining low-level edge cues with high-level semantic representations. Experimental results on five public datasets demonstrate that SGNet outperforms 16 state-of-the-art methods across six evaluation metrics, highlighting its superior segmentation performance and robustness.

1. Introduction

Colorectal cancer (CRC), recognized as the third most common global cancer (Bernal et al., 2012), predominantly originates from adenomatous polyps that undergo malignant progression over 10–15 years. Early detection and removal of these precancerous lesions, established as the most effective preventive measure (Favoriti et al., 2016), currently depend on colonoscopic screening. However, the accuracy of colonoscopy in diagnosis remains limited by subjective polyp demarcation, which exhibits clinically significant inter-observer variability. This critical limitation has driven the advancement of automated segmentation systems that deliver real-time objective lesion mapping, effectively standardizing diagnostic interpretation and therapeutic planning.

Recent advances in deep learning, especially Convolutional Neural Networks with powerful feature learning capabilities (Lecun et al., 1998), have significantly improved automated polyp segmentation. However, most existing methods still focus on local features, which limits their effectiveness in complex clinical scenarios. The advent of Transformer architectures (Vaswani et al., 2017) has recently promoted global context integration, complementing local features through comprehensive spatial-semantic analysis. Addressing persistent challenges like blurred backgrounds and ambiguous polyp boundaries necessitates

systematic integration of edge-texture features to advance segmentation accuracy.

Although existing methods have recently made progress in polyp segmentation, they still face difficulties in resolving blurred edges within polyp lesion regions. Edge blurring between polyps and adjacent tissues often stems from low contrast, high similarity, or gradual intensity transitions, all substantially compromising segmentation accuracy. Fig. 1 demonstrates three representative polyp cases with edge blurring and their segmentation implications. The first case exhibits polyp boundaries sharing chromatic similarity with surrounding tissues, creating low-contrast regions that challenge conventional edge detection algorithms. An irregular polyp morphology gradually merging with background tissue textures characterizes the second case, where high structural similarity complicates model discrimination. The third case displays progressive brightness and detail variations across transition zones, generating smooth indeterminate boundaries. Such edge ambiguity not hinders precise polyp localization but also elevates misclassification risks, ultimately limiting segmentation accuracy improvements.

To address these limitations, we propose a Selective Guidance Network with edge and texture awareness for polyp segmentation. The

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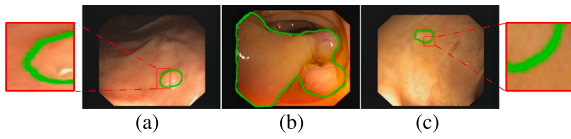


Fig. 1. Examples of polyps with blurred borders: (a) Low contrast boundary, (b) Irregular shape boundary, (c) Gradual brightness change boundary.

proposed architecture strategically extracts multi-scale edge and texture features and leverages them to refine the primary branch features through dense cross-level integration. The framework comprises four key components to ensure optimal performance. First, the Edge and Texture Awareness Module leverages multi-level fusion, Laplacian operators, and attention to enhance feature extraction. Second, the Multi-Scale Feature Enhancement Module removes redundant information while preserving essential prior knowledge. Third, the Prior Enhancement Module integrates edge-texture cues through cross-branch fusion, suppressing noise and refining feature representations. Finally, the Attention Gating Module performs intelligent multi-scale fusion across four stages, retaining the most informative features to improve segmentation accuracy.

Our main contributions are threefold as follows.

- We propose an Edge and Texture Awareness Module that extracts edge and texture features by fusing multi-scale features and combining the Laplacian operator with attention, which effectively resolves boundary prediction errors and improves segmentation.
- We design a Prior Enhancement Module that uses extracted edge and texture features to correct main branch features and enhances prior knowledge integration through feature cross-fusion.
- We introduce a Hierarchical Feature Fusion Module that filters noise with a four-layer attention mechanism and enables dense cross-level feature fusion, significantly boosting segmentation accuracy.

2. Related work

2.1. Polyp segmentation

Early polyp segmentation approaches depended on manually crafted features like color, shape, and texture (Van Wijk et al., 2010). These methods achieved detection and segmentation by extracting features and training classifiers. However, handcrafted features exhibited limited expressive power, often resulting in suboptimal segmentation outcomes (Park et al., 2012). Recent advancements in deep learning substantially advanced polyp segmentation techniques. Fully Convolutional Networks (FCNs) with dilated aggregation mechanisms were introduced by Sun et al. (2019) to learn high-level semantic representations. U-shaped architectures derived from U-Net (Ronneberger et al., 2015) subsequently became predominant. Optimized variants including ResUNet++ (Jha et al., 2019) and UNet++ (Zhou et al., 2019) exhibited enhanced segmentation capabilities. SANet (Wei et al., 2021) further improved feature extraction precision through a Shallow Attention Module and a Probability Correction Strategy (PCS), which the current work adopted to balance lesion-background region distributions. Transformer-based architectures further elevated polyp segmentation performance. Models like SwinUNet (Cao et al., 2022) and Polyp-PVT (Dong et al., 2023) exemplified this progression. DAEFormer (Azad et al., 2023) enhanced self-attention mechanisms by reconstructing spatial-channel relationships, whereas CAFE-Net (Liu et al., 2024) preserved low-level features via cross-attention and feature exploration networks. FMCA-Net (Li et al., 2025) addressed the boundary fuzziness problem in polyp segmentation by employing feature secondary reuse and a dilated convolution attention module. CP-SNet (Cai et al., 2025) tackled the challenge of segmenting polyps

with diverse shapes through multi-shape feature extraction and fine-grained region refinement. The proposed framework utilized Pyramid Vision Transformer (PVT) (Wang et al., 2021) as the encoder backbone in its U-shaped architecture. Building on these advancements, the proposed framework integrated the PVT as its encoder backbone and selectively incorporated prior knowledge to enhance feature representation, thereby directly addressing the persistent challenges of ambiguous boundaries encountered by existing methods in complex clinical scenarios.

2.2. Edge and texture awareness methods

Due to the irregular shapes and complex boundary characteristics of polyps, most existing methods focused on segmenting the overall region of polyps while neglected the importance of boundary information. As a result, accurately distinguishing the boundaries between polyps and surrounding tissues became challenging during segmentation, particularly at the edges, which compromised the precision and reliability of the results. To address this issue, Fang et al. (2019) proposed a boundary-sensitive loss function, which enhanced prediction accuracy by introducing region-boundary constraints. Psi-Net (Murugesan et al., 2019) integrated mask prediction with boundary feature extraction, effectively distinguishing the foreground (polyp) from the background. PraNet (Fan et al., 2020) introduced a parallel reverse attention network, and leveraged reverse attention modules to mine boundary cues and establish correlations between regions and boundaries. BT-Net (Li et al., 2024) enhanced the feature quality of the decoder by introducing transfer learning and a Boundary Feature Extraction module, effectively alleviating the segmentation challenges caused by blurred boundaries. EFA-Net (Zhou et al., 2025) improved polyp segmentation performance by leveraging cross-layer features to guide edge awareness. Texture features also played a crucial role in differentiating polyp foregrounds from backgrounds. TFCNet (Pan et al., 2024) proposed a Texture-aware Module and a Texture Feature Enhancement Module, which effectively fused low-level and high-level features. CFANet (Zhou et al., 2023) designed a boundary aggregation module to integrate boundary information with primary features, reducing noise and improving segmentation performance. MEGANet (Bui et al., 2024) combined classical edge detection techniques with attention mechanisms, effectively preserving high-frequency information, particularly edges and boundaries. Additionally, edge- and texture-aware methods were applied to camouflaged object recognition. For instance, FindNet (Li et al., 2022) enhanced prediction accuracy by combining global contour and local pattern information through Boundary Enhancement and Texture Enhancement modules. Inspired by these works, this study proposed a selective edge and texture awareness module and trained it using edge and texture labels, effectively acquiring prior knowledge to enhance segmentation accuracy.

2.3. Attention mechanism in feature fusion

Attention U-Net (Oktay et al., 2018) introduced a cost-efficient attention gating mechanism within its U-shaped architecture, enabling the model to prioritize critical information in the image and effectively guide cross-level feature fusion. SENet (Hu et al., 2018), on the other hand, incorporated a squeeze-and-excitation module combined with global average pooling to compute channel-wise recalibration, thereby enhancing key feature maps and reducing noise interference during feature fusion. Although channel attention mechanisms can identify feature maps requiring attention, they still exhibit limitations in precisely locating key regions within these feature maps. To address this issue, Chen et al. (2017) proposed a spatial attention module designed to more accurately focus on specific locations within feature maps. Further advancing this approach, Woo et al. (2018) introduced a convolutional block attention module that integrates both channel and spatial attention, enabling precise identification of important locations

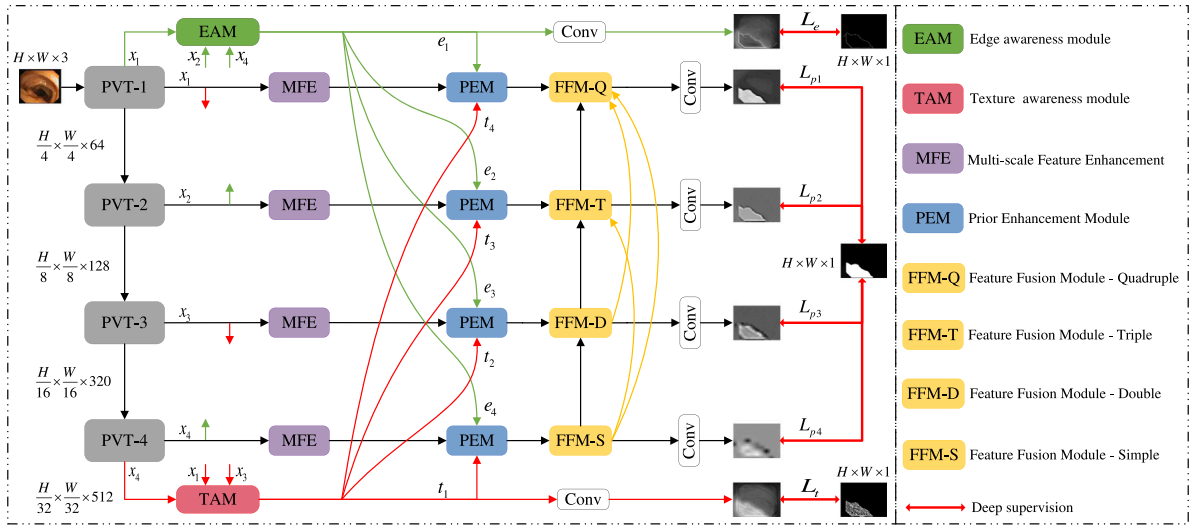


Fig. 2. Overview of the proposed SGNet.

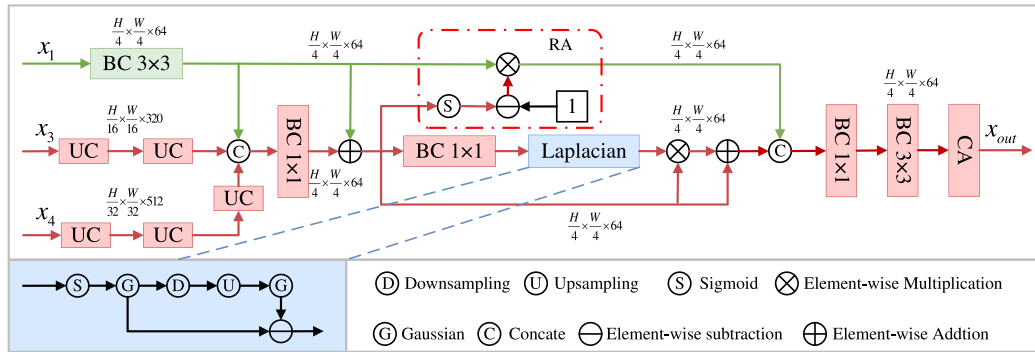


Fig. 3. Texture Awareness Module (TAM). UC refers to upsampling followed by convolution, BC represents a convolution block consisting of convolution, batch normalization (BN), and ReLU activation, and CA denotes coordinate attention.

and features in feature maps. This method has been widely adopted in multi-scale feature fusion. Hou et al. (2021) proposed a Coordinate Attention (CA) mechanism, which captures inter-channel interactions and incorporates direction-aware and position-sensitive properties, demonstrating exceptional capabilities in precisely focusing on target regions and capturing fine details. However, CA lacks a comprehensive consideration of cross-scale information during feature filtering and fusion, which may result in inadequate coordination between global and local information. To address this limitation, this study enhances the CA mechanism within the TAM and MFE modules by introducing a multi-scale feature enhancement strategy, thereby improving its capability to jointly model detailed and global semantic information. Furthermore, edge-aware characteristics are incorporated to optimize the attention distribution, leading to more effective feature filtering and fusion.

3. Method

3.1. Overview

This study proposes SGNet, a novel architecture based on the PVTv2 framework (Wang et al., 2022), designed to address polyp segmentation challenges by enhancing boundary clarity. The network extracts hierarchical features x_1 through x_4 , where x_1 preserves fine-grained spatial details, x_4 captures high-level semantics, and x_2 - x_3 integrate both detailed and contextual information. To tackle boundary ambiguity, a specialized multi-scale fusion strategy is implemented: larger-scale features emphasize boundary characteristics, while smaller-scale features

retain intricate textures. Edge-oriented features from x_1 , x_2 , and x_4 are selectively fused, whereas texture-sensitive features from x_1 , x_3 , and x_4 undergo targeted integration. An edge and texture awareness module further refines these representations under ground truth supervision, optimizing feature discrimination. After ETAM perceives the correct edge and texture features, these features need to be effectively integrated into the main features to enhance their expressiveness. First, the main features are filtered through MEFM to remove irrelevant information. Next, PEM is applied to correct the main features by utilizing the edge and texture features, further improving their effectiveness. Finally, after prior enhancement, multi-scale dense fusion is performed using HFFM to obtain the output features. The complete network architecture is shown in Fig. 2.

3.2. Edge and texture awareness module

Accurate edge and texture mapping is crucial for polyp segmentation based on backbone network features. While the multi-level features extracted by the backbone network effectively encode global image context, existing methods often struggle to simultaneously preserve high-frequency texture details and well-defined boundary features. To address this issue, we propose the edge and texture-aware module (ETAM). ETAM comprises EAM and TAM, responsible for capturing edge and texture features, respectively. The primary distinction between them lies in the different multi-scale features selectively fused. Fig. 3 illustrates the specific structure of TAM. The module follows a systematic three-stage design. First, preliminary multi-scale feature

fusion is performed. Then, features are extracted using the Laplacian operator and reverse attention mechanism. Finally, convolution and coordinate attention are applied to further refine edge and texture mapping, thereby enhancing segmentation accuracy.

For texture feature extraction in TAM, we fuse features from the x_1 , x_3 , and x_4 layers to ensure both fine detail preservation and global texture awareness. The x_1 layer captures fine-grained textures, compensating for the loss of detail at coarser scales. The x_3 layer encodes more complex patterns, enhancing the representation of advanced texture structures. The x_4 layer, in contrast, provides a broader semantic understanding of texture variations across the entire image. By integrating these multi-scale features, we achieve a comprehensive and precise texture representation. The extraction process is formulated as follows:

$$\begin{aligned} x'_1 &= BC_{3 \times 3}(x_1) \\ x'_3 &= UC(UC(x_3)) \\ x'_4 &= UC(UC(UC(x_4))) \\ x_{cat} &= CAT(x'_1, x'_3, x'_4) \end{aligned} \quad (1)$$

where $BC_{3 \times 3}$ represents a 3×3 convolution followed by BatchNorm and ReLU activation, UC denotes an upsampling operation followed by a 1×1 convolution, and CAT indicates feature concatenation. In contrast to TAM, the feature selection of EAM primarily focuses on low-level details. It integrates the features x_1 , x_2 , and x_4 , where x_1 captures local edge variations, x_2 enhances semantic consistency while preserving finer details, and x_4 provides global edge structure. This multi-scale fusion approach ensures a more robust and comprehensive edge representation. By selectively fusing these features, we improve both texture and edge representations, leading to enhanced segmentation performance. The effectiveness of this design is further validated by the ablation study.

To effectively distinguish between the foreground and background, we introduce a Background Attention Mechanism. This mechanism assigns different attention weights to each pixel, thereby enhancing the salient features of the foreground regions while suppressing background interference. The process can be mathematically described by the equation:

$$\begin{aligned} x'_{cat} &= BC_{1 \times 1}(x_{cat}) + x'_1 \\ x_b &= (1 - S(x'_{cat})) \times x'_1 \end{aligned} \quad (2)$$

where $BC_{1 \times 1}$ represents a 1×1 convolution followed by BatchNorm and ReLU activation applied to the concatenated feature map x_{cat} . S refers to the Sigmoid function, which scales the values in x'_{cat} to a range between 0 and 1, determining the attention weights. x'_1 is the refined feature map from earlier stages, and x_b represents the background features after applying the attention mechanism.

To further optimize segmentation accuracy, we integrate the Laplacian operator to capture local structural information, enhancing edge and detail features. The Laplacian operator helps in capturing subtle variations, particularly in boundary regions, thereby establishing a clearer distinction between the target object and the background. The Laplacian operation is defined as:

$$\begin{aligned} \text{Laplacian}(x) &= G(U(D(G(S(x)))) - G(S(x))) \\ x_l &= \text{Laplacian}(BC_{1 \times 1}(x'_{cat})) \times x'_{cat} + x'_{cat} \end{aligned} \quad (3)$$

where G represents Gaussian convolution, D denotes downsampling, U is upsampling, and S is the Sigmoid function. Next, we fuse the background attention features with Laplacian features to obtain the final feature representation. This process can be formulated as Eq. (4)

$$x_{lb} = CAT(x_b, x_l) \quad (4)$$

where CAT denotes concatenation, and x_b and x_l are the background attention and Laplacian features, respectively. After feature fusion, we further refine the features through a combination of convolution and attention mechanisms. First, convolution operations are applied to the fused features x_{lb} to perform smoothing and filtering, removing

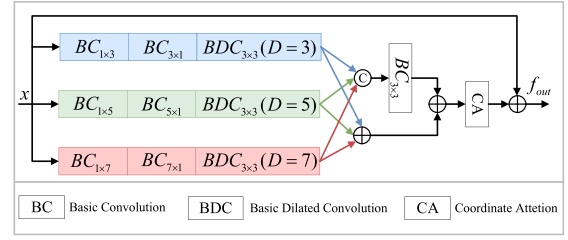


Fig. 4. Multi-Scale Feature Enhancement Module (MFE). $BC_{i \times j}$ denotes a standard convolution with a kernel size of $i \times j$, followed by Batch Normalization and ReLU activation. $BDC_{i \times j}$, $D = k$ denotes a dilated convolution with a kernel size of $i \times j$ and dilation rate k , also followed by BN and ReLU.

redundant information while enhancing the representation of critical features. Subsequently, an attention mechanism adaptively emphasizes key regions in the foreground, leading to improved recognition of target objects by the model. The synergy between convolution and attention mechanisms allows the model to better comprehend the relationship between foreground and background, resulting in the final output x_{out} . This process can be summarized as follows Eq. (5):

$$x_{out} = BC_{1 \times 1}(BC_{3 \times 3}(CA(x_{lb}))) \quad (5)$$

where CA denotes Coordinate Attention, $BC_{3 \times 3}$ refers to a 3×3 convolution followed by BatchNorm and ReLU, and $BC_{1 \times 1}$ represents a 1×1 convolution with BatchNorm and ReLU.

To guide the model in learning and extracting texture and edge features effectively, we optimize the Binary Cross-Entropy (BCE) loss function using texture and edge labels, thereby improving segmentation accuracy and refining the final detection results. Specifically, the texture label design is inspired by FindNet (Li et al., 2022), calculated as:

$$T_{map} = GT \times (\text{GrayImg} - B(\text{GrayImg})) \quad (6)$$

where GrayImg represents the input grayscale image, $B(\cdot)$ denotes the bilateral filtering operation, and GT is the ground truth of the polyp. Edge labels are derived from the CFANet (Zhou et al., 2023).

3.3. Prior enhancement module

Segmentation tasks face significant challenges in accurately predicting object boundaries and textures. First, boundary pixels are much rarer than foreground pixels, creating an imbalanced sample distribution that impairs the model's ability to effectively learn boundary features. Second, convolution and pooling operations in neural networks progressively reduce feature resolution, which can lead to the loss of the already sparse boundary and texture information. This issue is especially prevalent in polyp segmentation, as polyps often share similar colors with the intestinal background, while their diseased regions retain distinct texture features. These textures are crucial for precise polyp localization. To capture these features more effectively, we propose embedding edge and texture information into the extracted features, guiding the fusion process to improve segmentation performance. To ensure the extracted features are effective, we introduced a multi-scale feature enhancement module, as shown in Fig. 4. This module utilizes convolutional kernels of different sizes to extract features at multiple scales, capturing both fine-grained and global information. The multi-scale features are then fused through addition and concatenation, ensuring effective integration and minimizing redundancy and information loss.

Mathematically, given an input feature x , the features extracted through different convolutional kernels are formulated as follows:

$$\text{Branch}_k(x) = \begin{cases} BC_{1 \times 3}(BC_{3 \times 1}(BDC_{d=3}(x))), & k = s \\ BC_{1 \times 5}(BC_{5 \times 1}(BDC_{d=5}(x))), & k = m \\ BC_{1 \times 7}(BC_{7 \times 1}(BDC_{d=7}(x))), & k = l \end{cases} \quad (7)$$

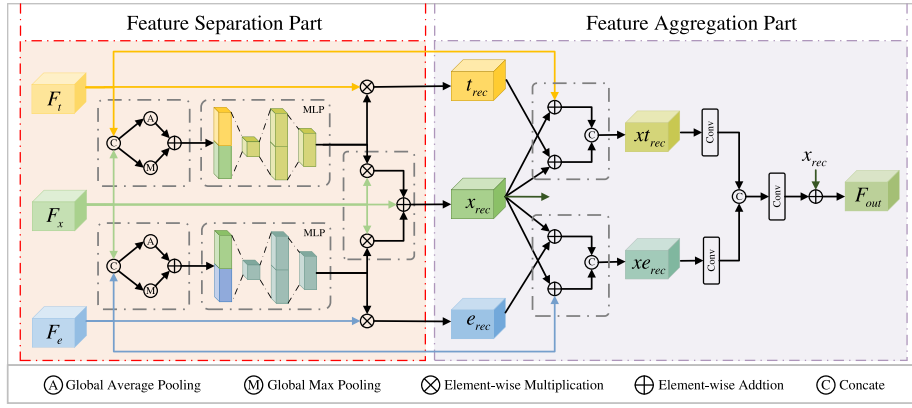


Fig. 5. Prior enhancement module (PEM).

where $BDC_d(x)$ represents the feature extraction operation using a convolutional kernel of size d , and $BC_{i \times j}$ denotes a convolution operation with a kernel size of $i \times j$. The fused features are then processed using addition and concatenation operations. It can be expressed as follows Eq. (8):

$$\begin{aligned} f_c &= \text{CAT}(\text{Branch}_s(x), \text{Branch}_m(x), \text{Branch}_l(x)) \\ f_a &= \text{Branch}_s(x) + \text{Branch}_m(x) + \text{Branch}_l(x) \end{aligned} \quad (8)$$

where Branch_s , Branch_m , and Branch_l denote branches with small, medium, and large receptive fields, respectively, used for extracting multi-scale features. The final output feature after attention-weighted fusion is Eq. (9):

$$f_{\text{out}} = \text{CA}(\text{BC}(f_c) + f_a) + x \quad (9)$$

Through this design, the model adaptively assigns different attention weights to different regions, highlighting key areas while suppressing irrelevant information. This enhanced representation provides a richer and more effective feature foundation for subsequent feature fusion and task learning. For the multimodal fusion module, we designed two core operations: feature recalibration for each modality and cross-modal feature aggregation. These operations optimize feature separation and feature aggregation, as illustrated in Fig. 5.

In the feature separation phase, we first fuse texture features F_t , primary features F_x , and edge features F_e pairwise to integrate information from different modalities effectively. To achieve this, we employ global average pooling and global max pooling operations, allowing the multi-level features to be compressed from both global and local perspectives. This process generates cross-modal attention vectors, which automatically focus on the important regions in each modality, thereby enhancing feature representation. It can be expressed as follows Eq. (10):

$$\begin{aligned} F_{xt} &= A(\text{CAT}(F_x, F_t)) + M(\text{CAT}(F_x, F_t)) \\ F_{xe} &= A(\text{CAT}(F_x, F_e)) + M(\text{CAT}(F_x, F_e)) \end{aligned} \quad (10)$$

where A represents global average pooling and M represents global max pooling. Next, the cross-modal attention vectors are mapped through a multi-layer perceptron (MLP) for non-linear transformation, further enhancing their expressiveness and discriminability. By introducing more non-linear relationships via MLP, the fusion between different modalities becomes more refined and informative.

Based on this recalibration, we apply weighted correction to each modality to ensure that their contributions to the final feature fusion are balanced, preventing any single modality from dominating the representation. This process can be described as Eq. (11):

$$\begin{aligned} t_{\text{rec}} &= \text{MLP}(F_{xt}) \times F_t \\ x_{\text{rec}} &= (\text{MLP}(F_{xt}) \times F_x) + (\text{MLP}(F_{xe}) \times F_x) \\ e_{\text{rec}} &= \text{MLP}(F_{xe}) \times F_e \end{aligned} \quad (11)$$

where MLP denotes a multi-layer perceptron, F_{xt} represents the interaction between the main features and texture features, and F_{xe} represents the combination of the main features and edge features. In the feature aggregation phase, the corrected features t_{rec} , x_{rec} , and e_{rec} are summed pairwise, and edge and texture features are incorporated as complementary information. This ensures that the complementary nature of different modalities is fully utilized. After multi-level progressive fusion, these features are passed through convolutional layers for further filtering and processing, ultimately extracting more precise segmentation features. F_{out} represents the final output feature, which is generated as follows:

$$\begin{aligned} xt_{\text{rec}} &= \text{CAT}((t_{\text{rec}} + x_{\text{rec}}), (x_{\text{rec}} + F_t)) \\ xe_{\text{rec}} &= \text{CAT}((e_{\text{rec}} + x_{\text{rec}}), (x_{\text{rec}} + F_e)) \\ F_{\text{out}} &= \text{conv}(\text{CAT}(\text{conv}(xt_{\text{rec}}), \text{conv}(xe_{\text{rec}}))) + x_{\text{rec}} \end{aligned} \quad (12)$$

where xt_{rec} and xe_{rec} denote the cross-fusion of the main features with texture features, and the main features with edge features, respectively. By applying this progressive optimization strategy of feature separation and aggregation, we effectively utilize prior knowledge to enhance the extracted features in an organic manner.

3.4. Hierarchical feature fusion module

To effectively preserve feature information and avoid redundancy, we designed a hierarchical feature fusion module in the cross-level feature fusion part. This module consists of four levels, aiming to improve feature representation through progressive feature enhancement and weighted fusion while maximizing the retention of key information. The implementation details are shown in Fig. 6, where p_i denotes the feature output of the i th layer.

Specifically, assuming that the input of the first level feature is f_m , it is processed through a multi-convolution attention-gated enhancement module. This module leverages different types of convolutions to capture multi-scale information, thereby enhancing feature diversity and improving the representation of fine-grained features. The primary goal of this stage is to extract local features and perform preliminary enhancement, resulting in feature p_4 . This process is represented by:

$$\begin{aligned} f_{mc} &= \text{CAT}(\text{PC}(f_m), \text{OC}(f_m), \text{DC}(f_m)) \\ p_4 &= S(\text{OC}(R(f_{mc}))) \times f_m + f_m \end{aligned} \quad (13)$$

where OC denotes ordinary convolution, PC denotes point convolution, DC denotes dilated convolution, S represents the sigmoid activation function, and R represents the ReLU activation function.

The subsequent levels further refine feature representations through hierarchical enhancement and attention mechanisms. At the second level, feature f_{l1} is first combined with the first-level output p_4 to generate an attention matrix, which captures the weight distribution of different features. The weighted features are then fused with the

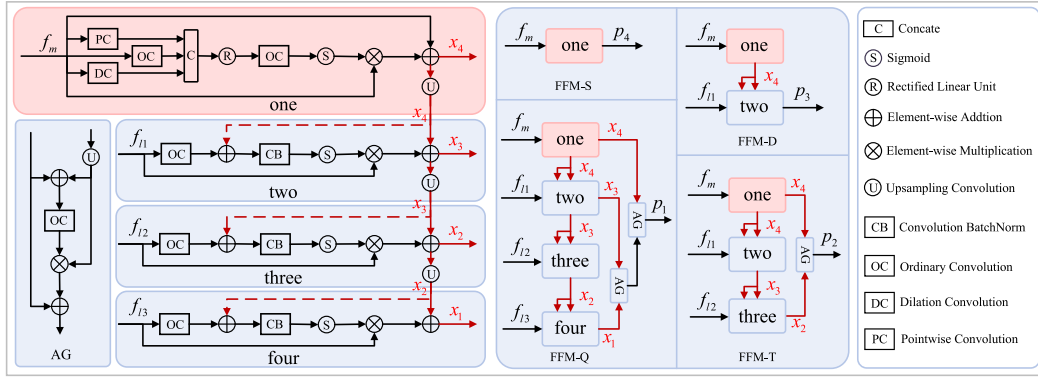


Fig. 6. Hierarchical feature fusion module (HFFM). The figure illustrates four feature fusion modules with different input configurations, progressively stacked from FFM-S to FFM-Q. Each module is built upon the previous one. One, Two, Three, and Four Represent the First to Fourth Levels of Gating.

original input, preserving fine details while enhancing feature discrimination through attention mechanisms. This results in feature p_3 :

$$\begin{aligned} p'_4 &= \text{UC}(p_4) \\ p_3 &= S(\text{CB}(\text{OC}(f_{i1}) + p'_4)) \times f_{i1} + p'_4 \end{aligned} \quad (14)$$

where UC denotes upsampling with convolution, and CB represents convolution followed by batch normalization. In the third layer, cross-layer fusion is further enhanced through an attention-gated mechanism. Specifically, the upsampled feature p_3 is merged with the third-layer input f_{i2} to align feature scales, thereby enabling effective fusion of representations from different levels, as described in Eq. (14). The resulting feature is then fused with the upsampled feature p_1 using the attention-gated mechanism to achieve dense connectivity and generate the feature p_2 . Finally, in the fourth layer, a cumulative fusion strategy is adopted to produce the final fused feature. The previously fused feature p_2 is first upsampled and combined with the fourth-layer input f_{i3} to retain the most informative cross-scale representations while suppressing redundant details. This result is then further fused with the upsampled features p_4 and p_3 , ultimately generating the enhanced feature p_1 , as also formulated in Eq. (14).

Through this step-by-step gated fusion mechanism, feature representations are progressively refined, ensuring robust information propagation across different levels and ultimately enhancing the segmentation performance.

3.5. Loss function

To effectively train the network, we adopted a combination of Binary Cross-Entropy Loss (BCE) and weighted Intersection over Union Loss (wIoU) as the segmentation training objective. The BCE loss offers robust convergence and accuracy for binary classification problems, while the wIoU effectively measures the overlap between the segmentation results and the ground truth, making it particularly suitable for images with blurry edges or complex structures. Based on this framework, we further incorporate edge and texture characteristics by applying separate BCE loss functions for edge and texture feature training. Our total loss function is defined as follows:

$$L_{\text{total}} = L_e + L_t + L_1 + L_2 + L_3 + L_4 \quad (15)$$

where L_e denotes the edge loss function, L_t represents the texture loss function, and L_i ($i = 1, 2, 3, 4$) corresponds to the loss function of the i th feature branch. Then, for the training of edge and texture features, we employ a simple Binary Cross-Entropy (BCE) loss function:

$$\begin{aligned} L_e &= \text{BCE}(G_e, e) \\ L_t &= \text{BCE}(G_t, t) \end{aligned} \quad (16)$$

where G_e denotes the ground truth for edge training, and e represents the predicted edge features. Similarly, G_t is the ground truth for texture

Table 1

Five overall polyp datasets and their division into training and testing datasets.

Datasets	Number	Train	Test
Kvasir-SEG	1000	900	100
CVC-ClinicDB	612	550	62
CVC-ColonDB	380	–	380
ETIS-LaribPolypDB	196	–	196
Endoscene	60	–	60

training, and t denotes the predicted texture features. For feature branch training, the segmentation loss is defined as:

$$L_i = L_{\text{seg}}(G_p, p_i) \quad (17)$$

where G_p represents the ground truth, and p_i denotes the output feature map from the i th layer.

$$L_{\text{seg}} = L_{\text{wIoU}} + L_{\text{wBCE}} \quad (18)$$

where L_{wIoU} and L_{wBCE} represent the Weighted IoU Loss (wIoU) and Weighted BCE Loss (wBCE), respectively, which are used for global and local constraints.

4. Experiments

4.1. Datasets

To evaluate the effectiveness of the proposed segmentation model, we conducted comparative experiments on five benchmark datasets. The detailed information for each dataset is as follows:

CVC-ClinicDB (Bernal et al., 2015): This dataset contains 612 images from colonoscopy video sequences.

ETIS-LaribPolypDB (Silva et al., 2014): This dataset consists of 196 polyp images.

CVC-ColonDB (Bernal et al., 2012): This dataset includes 380 colorectal polyp images.

Kvasir (Jha et al., 2020): This dataset contains 1000 colorectal polyp images collected from multiple colonoscopy video sequences.

Endoscene (corresponding to CVC-300) (Vázquez et al., 2017): This dataset contains 60 colorectal polyp images.

We use the same data distribution setting as PraNet (Fan et al., 2020) on the five datasets (Kvasir-SEG, CVC-ClinicDB, CVC-ColonDB, ETIS-LaribPolypDB, and Endoscene). Specifically, images from Kvasir-SEG and CVC-ClinicDB are randomly split, with 90% used for training and 10% for testing. Additionally, we assess the model's generalization ability on the unseen CVC-ColonDB, ETIS-LaribPolypDB, and EndoScene datasets. The division of the five datasets into training and testing datasets is shown in Table 1.

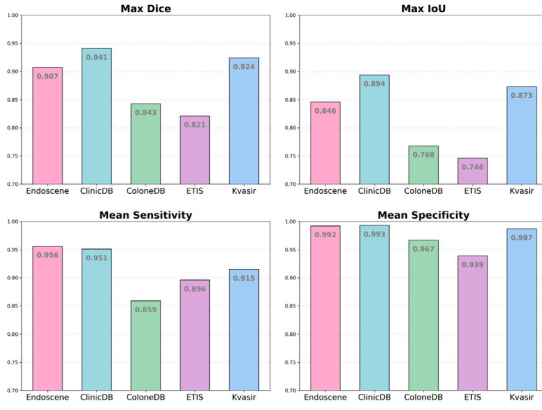


Fig. 7. Four additional metrics across the five polyp datasets.

4.2. Evaluation metrics

We use eight evaluation metrics to assess the model's performance: Dice coefficient (Milletari et al., 2016), Intersection over Union (IoU), Mean Absolute Error (MAE) (Perazzi et al., 2012), weighted F-measure (F_{β}^w) (Achanta et al., 2009), S-measure (S_{α}) (Fan et al., 2017), max E-measure ($\max E_{\varepsilon}$) (Fan et al., 2018), sensitivity, and specificity.

We primarily focus on the Dice coefficient and Intersection over Union (IoU) as key evaluation metrics. The Dice coefficient quantifies the similarity between the predicted and ground truth regions by computing the ratio of twice their intersection to the sum of their individual areas. In contrast, IoU measures the overlap between the predicted and ground truth regions by dividing the intersection by their union. In addition, the Mean Absolute Error (MAE) evaluates the pixel-wise absolute differences by averaging them over all pixels, providing an overall measure of prediction accuracy. To address the limitations of pixel-wise metrics, the weighted F-measure (F_{β}^w) balances precision and recall with greater emphasis on object-level consistency. The structure measure (S_{α}) captures structural similarity by integrating object-aware and region-aware information. The maximum enhanced-alignment measure ($\max E_{\varepsilon}$) assesses segmentation quality from both global and local perspectives. Finally, sensitivity and specificity quantify the proportions of true positives and true negatives, respectively, offering insight into the model's ability to differentiate between foreground and background regions.

4.3. Implementation details

We implemented SGNet using the PyTorch framework and accelerated the computation with a computer equipped with an NVIDIA RTX 4090 (24 GB) GPU. Given the variation in the size of each polyp image, we adopted a multi-scale strategy (Fan et al., 2020) during training. The specific hyperparameter settings are as follows: The network parameters are updated using the AdamW optimizer (Loshchilov & Hutter, 2017), with a learning rate of 1×10^{-4} and weight decay set to 1×10^{-4} . Additionally, the applied data augmentation techniques include random flipping, random rotation, random cropping, randomPepper noise, and color jittering, which enhance the model's generalization ability and effectively mitigate the risk of overfitting. The input image size is set to 352×352 , the mini-batch size is 16, and the training process spans 150 epochs.

4.4. Performance comparisons

4.4.1. Comparison methods

To evaluate the effectiveness of the proposed polyp separation method, we compared it with sixteen state-of-the-art methods, including U-Net (Ronneberger et al., 2015), U-Net++ (Zhou et al., 2019),

PraNet (Fan et al., 2020), SANet (Wei et al., 2021), PolypPVT (Dong et al., 2023), CFA-Net (Zhou et al., 2023), M2SNet (Zhao et al., 2023), TMF-Net (Yang et al., 2023), PVT-Cascade (Rahman & Marculescu, 2023), CSCAU-Net (Shu et al., 2024), NPD-Net (Yu et al., 2024), UHA-Net (Zhou et al., 2024), BR-Net (Yue et al., 2024), MEGANet (Bui et al., 2024), and EMTS-Net (Wang et al., 2024), SAFE-Net (Yu & Qi, 2025). The comparison results are presented in Tables 2 and 3.

4.4.2. Quantitative comparison

The evaluation framework employs five benchmark datasets to comprehensively assess model performance. Kvasir-SEG and CVC-ClinicDB serve to evaluate the capacity for feature modeling, whereas the other three datasets (CVC-ColonDB, ETIS-LaribPolypDB, and Endoscene) assess generalization capabilities across diverse clinical scenarios.

Table 2 quantitatively compares our approach with sixteen existing segmentation methods across three challenging datasets. The data for all compared methods are directly sourced from the original papers. The experimental results reveal consistent performance advantages of our model over conventional and modern segmentation techniques. Particularly significant enhancements emerge in cross-dataset evaluations, where the proposed method achieves 5.9%, 5.9%, 6.4%, 4.1%, 3.2%, and 1.4% metric improvements compared to MEGANet on CVC-ColonDB. The ETIS-LaribPolypDB evaluation shows respective gains of 2.8%, 3.2%, 2.9%, 2.1%, 2.1%, 0.9%, and 0.1% across evaluation metrics. For the CVC-300 benchmark, performance improvements reach 1.6%, 2.4%, 2.5%, 1.6%, 1.1%, and 0.4%. These consistent advancements across multiple independent datasets confirm the model's robust generalization capacity in varying endoscopic imaging conditions.

Table 3 compares the performance of our model with sixteen state-of-the-art methods on six metrics across the CVC-ClinicDB and Kvasir datasets, using data directly sourced from the original papers of these methods. On the CVC-ClinicDB dataset, our model outperforms all other methods in most metrics. Although SAFE-Net and EMTS-Net show strong segmentation performance, and MEGANet considers the dependencies between regions and boundaries, none of these methods can effectively segment polyps with blurred edges. In contrast, our model demonstrates the ability to accurately segment polyps and achieve superior performance. This is attributed to the model's capability to capture multi-scale information, addressing the scale variation of polyps, and its utilization of boundary and texture-aware features, which provide more detailed boundary cues to enhance segmentation performance. On the Kvasir dataset, our proposed polyp segmentation method consistently achieves the best performance. Specifically, compared to MEGANet, our model demonstrates improvements of 1.0%, 1.1%, 1.3%, 1.1%, 1.1%, and 0.5% in terms of mDice, mIoU, F_{β}^w , S_{α} , $\max E_{\varepsilon}$, and MAE, respectively.

In addition to the overall quantitative comparisons presented in the above tables, we further provide precision-recall and F-measure curves in Figs. 9 and 10. The results clearly show that our proposed model outperforms other state-of-the-art polyp segmentation methods. Moreover, in Fig. 7, we supplement the bar graph with the four metrics of our model on five datasets, further demonstrating the excellent performance achieved.

4.4.3. Qualitative comparison

Fig. 8 presents the segmentation results of our model compared to nine state-of-the-art polyp segmentation methods. The visual results clearly show that our method closely approximates the ground truth segmentation map and outperforms the other methods in handling various challenging factors. Specifically, in rows 2, 4, 7, and 9, the polyps are very small, but our model can still accurately segment them through the prior enhancement of edge and texture features. In contrast, UNet and UNet++ fail to segment the small polyps successfully. Other methods, such as NPD-Net, MEGANet, CSCAU-Net, and PraNet, produce multiple over-segmentation errors. While Cascade, CFA-Net, and PolypPvt achieve relatively accurate segmentation, they fail to

Table 2

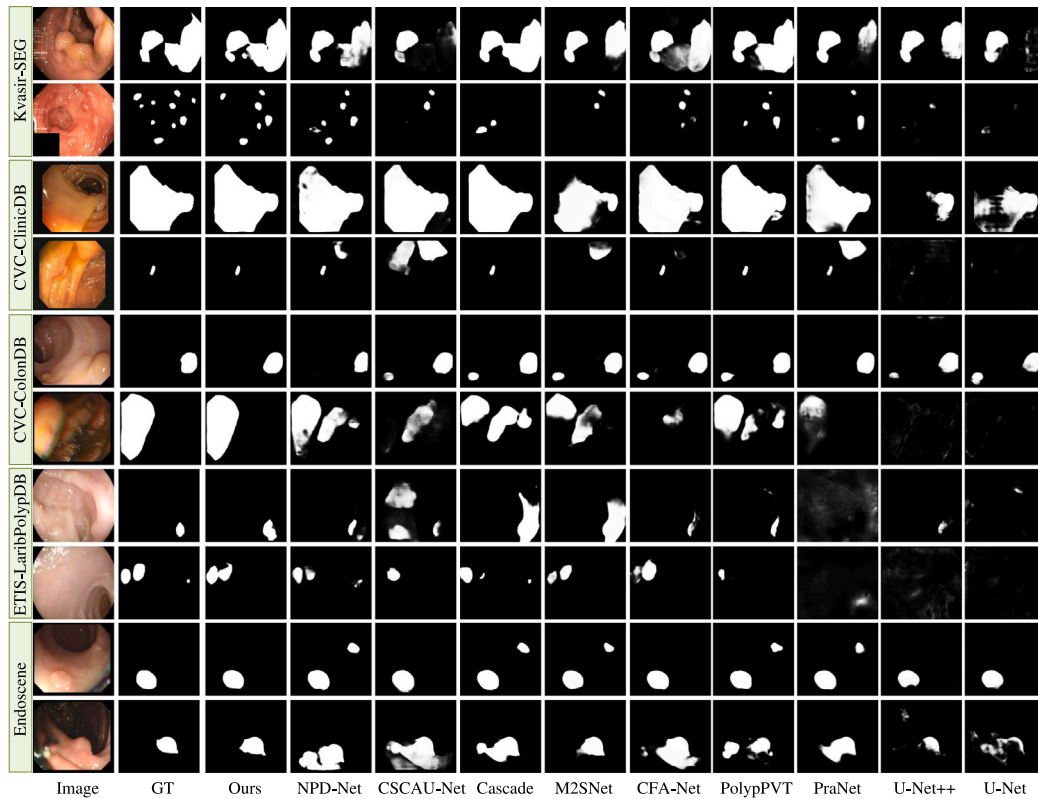
The experimental comparison of different network models on ETIS-LaribPolypDB, CVC-ColonDB, and Endoscene datasets.

Methods	Year	CVC-ColonDB						ETIS-LaribPolypDB						Endoscene					
		mDice $\uparrow$	mIoU $\uparrow$	$F_{\beta}^w \uparrow$	$S_{\alpha} \uparrow$	$\max E_{\xi} \uparrow$	MAE $\downarrow$	mDice $\uparrow$	mIoU $\uparrow$	$F_{\beta}^w \uparrow$	$S_{\alpha} \uparrow$	$\max E_{\xi} \uparrow$	MAE $\downarrow$	mDice $\uparrow$	mIoU $\uparrow$	$F_{\beta}^w \uparrow$	$S_{\alpha} \uparrow$	$\max E_{\xi} \uparrow$	MAE $\downarrow$
U-Net	2015	0.512	0.444	0.498	0.712	0.776	0.061	0.398	0.335	0.366	0.684	0.740	0.036	0.710	0.627	0.684	0.843	0.876	0.022
U-Net++	2018	0.483	0.410	0.467	0.691	0.760	0.064	0.401	0.344	0.390	0.683	0.776	0.035	0.707	0.624	0.687	0.839	0.898	0.018
PraNet	2020	0.712	0.640	0.699	0.820	0.872	0.043	0.628	0.567	0.600	0.794	0.841	0.031	0.871	0.797	0.843	0.925	0.972	0.010
SA-Net	2021	0.745	0.665	0.726	0.834	0.879	0.041	0.761	0.670	0.703	0.859	0.904	0.015	0.872	0.792	0.834	0.912	0.966	0.010
PolypPVT	2023	0.808	0.727	0.795	0.865	0.919	0.031	0.787	0.706	0.750	0.871	0.910	0.013	0.900	0.833	0.884	0.935	0.973	0.007
CFA-Net	2023	0.743	0.665	0.728	0.835	0.898	0.039	0.732	0.655	0.693	0.845	0.892	0.014	0.893	0.827	0.875	0.938	0.978	0.008
M2SNet	2023	0.758	0.685	0.737	0.842	0.869	0.038	0.749	0.678	0.712	0.846	0.872	0.017	0.903	0.842	0.881	0.939	0.965	0.009
TMF-Net	2023	0.709	0.627	0.693	0.814	0.871	0.040	0.651	0.565	0.606	0.796	0.857	0.025	0.878	0.803	0.848	0.925	0.973	0.008
PVT-Cascade	2023	0.789	0.710	0.779	0.855	0.901	0.031	0.786	0.712	0.759	0.872	0.902	0.013	0.892	0.824	0.873	0.932	0.966	0.009
CSCAU-Net	2024	0.788	0.703	0.760	0.857	0.906	0.036	0.688	0.608	0.644	0.814	0.882	0.026	0.868	0.789	0.835	0.920	0.969	0.012
NPD-Net	2024	0.812	0.729	0.795	0.869	0.923	0.029	0.782	0.700	0.739	0.872	0.918	0.017	0.876	0.803	0.848	0.925	0.971	0.011
UHA-Net	2024	0.769	0.695	0.755	0.849	0.892	0.036	0.746	0.670	0.709	0.854	0.905	0.023	0.902	0.838	0.886	0.942	0.976	0.006
BRNet	2024	0.795	0.723	0.776	0.866	0.892	0.031	0.760	0.696	0.716	0.857	0.859	0.017	0.898	0.836	0.878	0.941	0.956	0.007
MEGANet	2024	0.781	0.706	0.766	0.845	0.899	0.038	0.789	0.709	0.753	0.866	0.915	0.015	0.887	0.818	0.863	0.924	0.959	0.009
EMTS-Net	2024	0.788	0.708	0.770	0.852	0.898	0.037	0.780	0.702	0.737	0.869	0.905	0.016	0.900	0.833	0.882	0.935	0.977	0.006
SAFE-Net	2025	0.781	0.704	0.767	0.858	0.908	0.034	0.794	0.713	0.748	0.884	0.919	0.015	0.895	0.828	0.872	0.938	0.978	0.008
SG-Net(ours)	2025	0.840	0.765	0.829	0.886	0.931	0.024	0.817	0.741	0.782	0.887	0.924	0.014	0.903	0.842	0.888	0.940	0.970	0.005

Table 3

The experimental comparison of different network models on Kvasir-SEG and CVC-ClinicDB datasets.

Methods	Year	Kvasir-SEG						CVC-ClinicDB					
		mDice $\uparrow$	mIoU $\uparrow$	$F_{\beta}^w \uparrow$	$S_{\alpha} \uparrow$	$\max E_{\xi} \uparrow$	MAE $\downarrow$	mDice $\uparrow$	mIoU $\uparrow$	$F_{\beta}^w \uparrow$	$S_{\alpha} \uparrow$	$\max E_{\xi} \uparrow$	MAE $\downarrow$
U-Net	2015	0.818	0.746	0.794	0.858	0.893	0.055	0.823	0.755	0.811	0.889	0.954	0.019
U-Net++	2018	0.821	0.743	0.808	0.862	0.910	0.048	0.794	0.729	0.785	0.873	0.931	0.022
PraNet	2020	0.894	0.836	0.880	0.912	0.944	0.030	0.899	0.849	0.896	0.936	0.979	0.009
SA-Net	2021	0.897	0.839	0.883	0.910	0.946	0.031	0.911	0.853	0.903	0.939	0.973	0.011
PolypPVT	2023	0.917	0.864	0.911	0.925	0.962	0.023	0.937	0.889	0.936	0.949	0.989	0.006
CFA-Net	2023	0.915	0.861	0.903	0.924	0.962	0.023	0.933	0.883	0.924	0.950	0.989	0.007
M2SNet	2023	0.912	0.861	0.901	0.922	0.953	0.025	0.922	0.880	0.917	0.942	0.970	0.009
TMF-Net	2023	0.872	0.806	0.852	0.893	0.933	0.035	0.892	0.839	0.886	0.928	0.964	0.011
PVT-Cascade	2023	0.911	0.863	0.906	0.919	0.964	0.025	0.919	0.872	0.918	0.936	0.973	0.013
CSCAU-Net	2024	0.903	0.846	0.890	0.918	0.951	0.031	0.915	0.864	0.912	0.942	0.975	0.010
NPD-Net	2024	0.910	0.855	0.899	0.921	0.958	0.026	0.928	0.878	0.926	0.946	0.982	0.008
UHA-Net	2024	0.908	0.857	0.901	0.926	0.955	0.026	0.927	0.881	0.923	0.953	0.979	0.007
BRNet	2024	0.918	0.871	0.905	0.928	0.957	0.026	0.921	0.878	0.912	0.948	0.967	0.008
MEGANet	2024	0.911	0.859	0.904	0.916	0.954	0.026	0.930	0.885	0.931	0.950	0.980	0.008
EMTS-Net	2024	0.919	0.869	0.912	0.927	0.962	0.022	0.935	0.885	0.932	0.949	0.988	0.008
SAFE-Net	2025	0.917	0.865	0.908	0.931	0.963	0.025	0.937	0.889	0.931	0.958	0.991	0.007
SG-Net(ours)	2025	0.921	0.870	0.917	0.927	0.965	0.021	0.938	0.892	0.937	0.950	0.987	0.006

**Fig. 8.** Qualitative visualization of polyp segmentation results comparing our model with nine state-of-the-art methods.

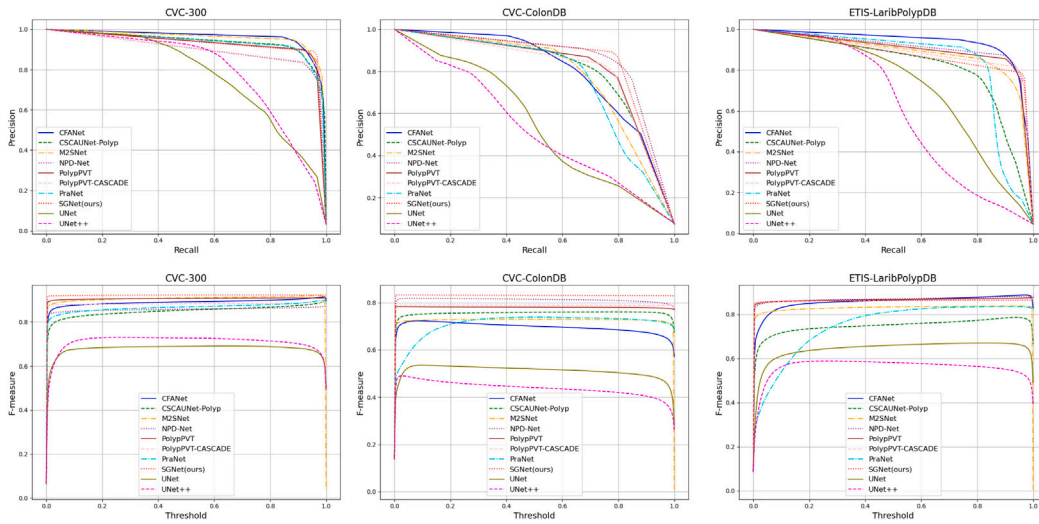


Fig. 9. Precision-Recall and F-measure curves of our model and other nine state-of-the-art methods across three datasets.

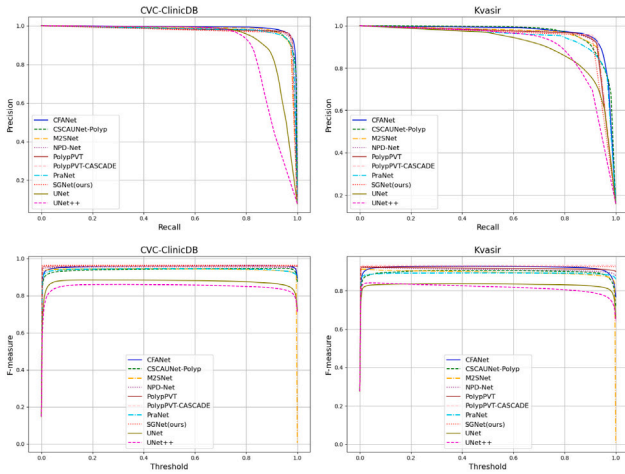


Fig. 10. Precision-Recall and F-measure curves of our model and other nine state-of-the-art methods across two datasets.

capture the entire polyp, leaving some regions unsegmented. In rows 1, 3, 6, and 10, the polyps are larger and exhibit varying shapes, which pose greater challenges for accurate segmentation. As a result, UNet, UNet++, M2SNet, and PraNet show inferior performance compared to the other methods. In rows 5 and 8, the boundary between the polyp and the background is unclear, causing the polyp and background to visually blend together. Despite this, our method is able to segment the polyps more accurately, outperforming other methods in these challenging scenarios. Overall, the visual results further confirm the superior performance of our model in addressing a wide range of challenges in polyp segmentation, particularly when dealing with images that feature unclear boundaries. These results demonstrate the robustness and accuracy of our model in complex segmentation tasks.

4.5. Ablation study

4.5.1. Effectiveness of different parts of sgnet

In this study, we introduced five novel modules, namely EAM, TAM, PEM, MFE, and HFFM, to address the challenge of polyp edge differentiation. To assess the effectiveness of each module, we conducted eight sets of ablation experiments on five datasets. The results are summarized in Table 4.

We first evaluated the individual contributions of MFE and HFFM. As shown in rows 1 and 2 of Table 4, neither module alone achieved optimal segmentation performance, though HFFM demonstrated a slight advantage over MFE. Notably, when both modules were combined (row 3), their synergy led to improved performance, underscoring their complementary nature. Next, we examined the impact of incorporating the EAM module. The results indicate that adding EAM improved the mDice index by 0.3%, 0.9%, 0.9%, and 0.5% on the CVC-ClinicDB, Kvasir, Endoscene, and CVC-ColonDB datasets, respectively. However, on the ETIS-LaribPolypDB dataset, mDice decreased by 0.9%, suggesting that while EAM is crucial for polyp segmentation in most cases, its effectiveness may vary depending on dataset characteristics. We then assessed the addition of the TAM module (row 5). Compared to using only EAM, this modification led to a further 0.1% increase in mDice on the CVC-ClinicDB dataset, but performance declined on the remaining four datasets. This suggests that, without explicit boundary guidance from EAM, relying solely on TAM is insufficient for accurate polyp segmentation. Rows 6 and 7 present results for models incorporating the PEM module alongside EAM and TAM. In row 6, MFE was introduced into the framework, which strengthened boundary delineation. However, in the absence of HFFM, excessive information loss occurred during cross-level feature fusion, preventing further performance gains. In contrast, the experiment in row 7 replaced MFE with HFFM. While this resulted in some loss of multi-scale information, the dense feature fusion capability of HFFM still enabled notable segmentation improvements. Finally, row 8 represents our complete model, SGNet, which achieved the highest mDice scores on three datasets and ranked second on the other two. These findings further validate the complementary and synergistic effects of the proposed modules, demonstrating that our method significantly enhances polyp segmentation performance.

Additionally, as illustrated in Fig. 13, we visualize the heatmaps of P1 features across the eight ablation experiments, with model numbers corresponding to Table 4. The following observations can be made: in model 4, the inclusion of EAM significantly improves edge segmentation accuracy. In model 5, the introduction of TAM enhances texture feature extraction; however, edge delineation remains suboptimal. In contrast, model 8, which integrates both EAM and TAM, effectively focuses on the lesion region and accurately outlines its boundaries. Models 1, 2, and 3, lacking key modules, tend to emphasize non-lesion areas, leading to reduced segmentation performance. Moreover, while model 7 successfully highlights lesion regions, it retains a degree of residual noise, highlighting the importance of effective feature filtering before fusion.

Table 4

Segmentation results of ablation experiments on various modules of our model.

Components					ClinicDB		Kvasir-SEG		Endoscene		ColonDB		ETIS	
EAM	TAM	PEM	MFE	HFFM	mDice	mIoU	mDice	mIoU	mDice	mIoU	mDice	mIoU	mDice	mIoU
			✓		0.932	0.883	0.915	0.866	0.893	0.829	0.805	0.726	0.802	0.718
				✓	0.934	0.888	0.912	0.860	0.904	0.839	0.816	0.739	0.797	0.720
			✓	✓	0.935	0.887	0.911	0.857	0.891	0.825	0.828	0.752	0.800	0.720
✓			✓	✓	0.938	0.893	0.920	0.869	0.900	0.838	0.833	0.756	0.791	0.710
			✓	✓	0.939	0.893	0.913	0.862	0.882	0.815	0.819	0.741	0.789	0.703
✓	✓		✓	✓	0.934	0.884	0.909	0.855	0.900	0.838	0.812	0.736	0.815	0.745
✓	✓	✓		✓	0.937	0.891	0.920	0.869	0.904	0.839	0.818	0.739	0.806	0.732
✓	✓	✓	✓	✓	0.938	0.892	0.921	0.870	0.903	0.842	0.840	0.765	0.817	0.741

Table 5

Comparison of segmentation results for the ablation study on multi-level feature selection in the EAM module.

Components				ClinicDB		Kvasir-SEG		Endoscene		ColonDB		ETIS	
1	2	3	4	mDice	mIoU	mDice	mIoU	mDice	mIoU	mDice	mIoU	mDice	mIoU
✓				0.928	0.876	0.907	0.848	0.893	0.828	0.819	0.740	0.823	0.746
✓			✓	0.936	0.889	0.907	0.853	0.901	0.838	0.824	0.749	0.818	0.745
✓	✓		✓	0.938	0.892	0.921	0.870	0.903	0.842	0.840	0.765	0.817	0.741
✓	✓	✓	✓	0.940	0.894	0.908	0.854	0.895	0.831	0.819	0.744	0.805	0.732

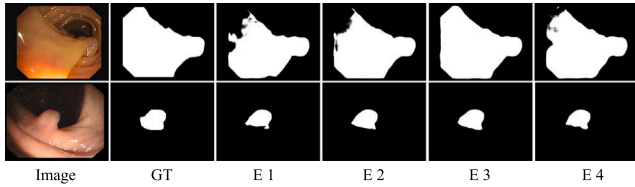
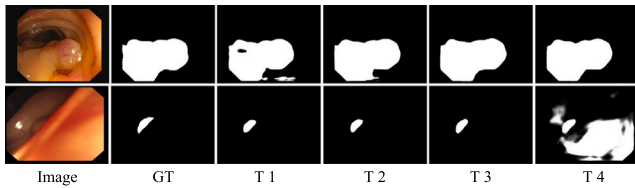
Table 6

Comparison of segmentation results for the ablation study on multi-level feature selection in the TAM module.

Components				ClinicDB		Kvasir-SEG		Endoscene		ColonDB		ETIS	
1	2	3	4	mDice	mIoU	mDice	mIoU	mDice	mIoU	mDice	mIoU	mDice	mIoU
✓				0.933	0.883	0.905	0.853	0.912	0.852	0.822	0.746	0.819	0.747
✓			✓	0.938	0.891	0.910	0.857	0.901	0.837	0.829	0.752	0.809	0.738
✓		✓	✓	0.938	0.892	0.921	0.870	0.903	0.842	0.840	0.765	0.817	0.741
✓	✓	✓	✓	0.932	0.883	0.909	0.856	0.891	0.827	0.820	0.747	0.796	0.724

Table 7Ablation study on swapping feature combinations between EAM and TAM. **Swap** denotes exchanging the selected feature inputs of EAM and TAM.

Methods	ClinicDB		Kvasir-SEG		Endoscene		ColonDB		ETIS	
	mDice	mIoU	mDice	mIoU	mDice	mIoU	mDice	mIoU	mDice	mIoU
Swap	0.939	0.893	0.918	0.868	0.900	0.837	0.836	0.765	0.795	0.715
Ours	0.938	0.892	0.921	0.870	0.903	0.842	0.840	0.765	0.817	0.741

**Fig. 11.** Visual comparison is used to verify the advantages of feature fusion at different levels of EAM.**Fig. 12.** Visual comparison is used to verify the advantages of feature fusion at different levels of TAM.

4.5.2. Effectiveness of boundary-texture aware features

In order to validate the effectiveness of selectively integrating multi-level features to enhance edge and texture perception, we conducted four sets of ablation experiments on the Edge Awareness Module (EAM) and the Texture Awareness Module (TAM). These experiments were designed to compare the impact of different feature combinations within each module. The segmentation results are reported in [Tables 5 and 6](#), while the corresponding visualizations are presented in [Figs. 11 and 12](#).

Additionally, [Table 7](#) provides the results of an ablation study in which the feature combinations assigned to EAM and TAM were swapped, in order to further investigate the effect of module arrangement on overall performance.

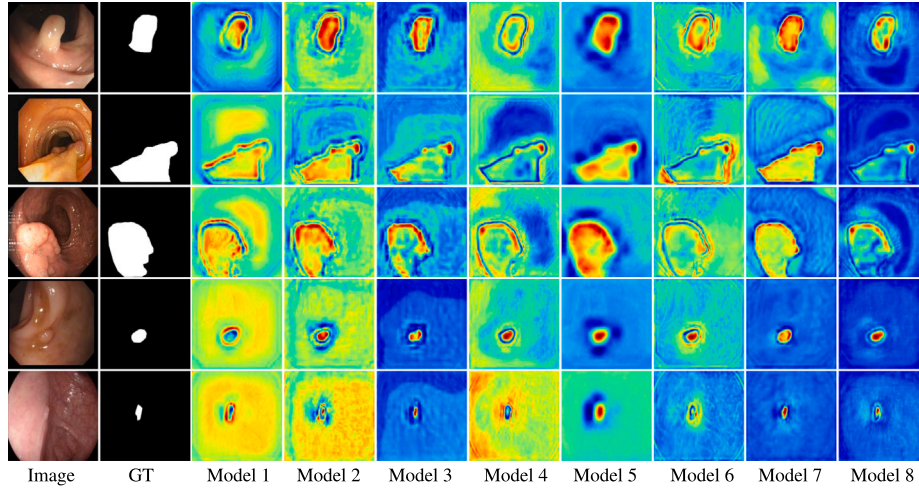
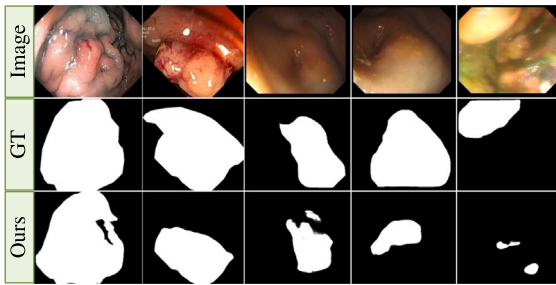
The experimental results reveal that different feature layers contribute distinct types of perceptual information. The first-layer feature (x_1) captures the richest spatial details and edge cues, making it well-suited for fine-grained boundary modeling. In contrast, the deepest-layer feature (x_4) contains the most abstract global semantic information, which is essential for interpreting complex textures and high-level structures. Intermediate-layer features (x_2 and x_3) serve as bridges between local detail and semantic abstraction. Their respective biases influence whether the model emphasizes precise boundary localization or robust texture representation.

Based on this hierarchical analysis, we designed a selective cross-level feature integration strategy tailored for each module. For the EAM, we selected shallow features (x_1, x_2) to reinforce edge detail preservation, while incorporating deep semantic features (x_4) to guide accurate boundary alignment. In contrast, the TAM emphasizes mid-to-deep semantic features (x_3, x_4) to enhance texture modeling, while retaining the low-level feature x_1 to preserve spatial granularity in texture patterns. It is worth highlighting that in the fourth experimental setting, where all features (x_1, x_2, x_3, x_4) were integrated without distinction, the model's performance declined despite the increased theoretical richness of information. This is mainly due to feature redundancy — particularly from the intermediate layers — which introduces unnecessary noise and increases computational cost. In contrast, our proposed selective integration strategy strikes a better balance between performance and efficiency, underscoring the value of deliberate feature selection in modular design.

Table 8

Ablation experiment comparison of ETAM with other network edge and texture related modules.

Methods	ClinicDB		Kvasir-SEG		Endoscene		ColonDB		ETIS	
	mDice	mIoU	mDice	mIoU	mDice	mIoU	mDice	mIoU	mDice	mIoU
TAM (Pan et al., 2024)	0.938	0.891	0.913	0.861	0.907	0.847	0.828	0.754	0.793	0.720
BFE (Li et al., 2024)	0.939	0.894	0.912	0.860	0.893	0.828	0.830	0.752	0.816	0.740
EGM (Zhou et al., 2025)	0.940	0.893	0.900	0.847	0.895	0.834	0.812	0.735	0.807	0.730
ETAM (ours)	0.938	0.892	0.921	0.870	0.903	0.842	0.840	0.765	0.817	0.741

**Fig. 13.** Heatmap visualization of segmentation results from the ablation study of our model.**Fig. 14.** Failure cases in challenging scenes.

Furthermore, to further validate the superiority of ETAM, we conducted comparative ablation studies with edge and texture related modules from other networks. As shown in Table 8, we replaced our TAM module with the TAM module from TFC-Net (Pan et al., 2024), and substituted our EAM module with the BFE module from BT-Net (Li et al., 2024) and the EGM module from EFA-Net (Zhou et al., 2025). The results show that our ETAM module achieved the best performance among all compared configurations. This improvement can be attributed to our selective fusion of multi-level features, which enhances edge and texture perception. Other models do not fully address this aspect. In addition, the integration of the Laplacian operator with attention mechanisms enables more precise feature refinement. The combination of structural priors and adaptive attention is a key factor contributing to the effectiveness of our method.

4.6. Failure cases

As shown in Fig. 14, although the proposed SGNet demonstrates progress in boundary differentiation, it still presents certain limitations when addressing cases with large polyp regions and complex internal structures. For instance, in the first column of Fig. 14, while the model identifies the boundary relatively accurately, it struggles to capture the

detailed internal features of the polyp. The subsequent columns present additional failure cases in which the polyps are large and contain intricate textures, leading SGNet to achieve only partial recognition or, in some instances, fail to detect the target area altogether. This limitation may stem from the model's focus on boundary information during training, which can reduce its sensitivity to internal texture variations. Similar issues are also observed in existing polyp segmentation methods such as BR-Net (Yue et al., 2024), MEGANet (Bui et al., 2024), and EMTS-Net (Wang et al., 2024). Future work will aim to develop more effective strategies for balancing edge and texture information, particularly in challenging scenarios involving large and structurally complex polyps, thereby enhancing the overall segmentation performance.

5. Conclusion

In this paper, we proposed an Edge and Texture Awareness Selectively Guided Network for polyp segmentation. First, we designed an Edge and Texture Awareness Module, which aimed to accurately capture edge and texture features, thereby improving the precision of segmentation boundaries. To more effectively utilize these perceptual features, we introduced a Prior Enhancement Module that fused edge and semantic information into the main features, enhancing the model's ability to distinguish between polyps and background. To address the scale variation problem and fully leverage hierarchical features, we presented a Hierarchical Feature Fusion Module, which improved segmentation performance at different scales by fusing cross-layer features and integrating multi-scale contextual information. Additionally, we proposed a Multi-Scale Feature Enhancement Module, which enabled the model to focus on more distinguishable feature regions by removing noise from the main features. Experimental results on five public datasets demonstrated that SG-Net outperformed existing state-of-the-art methods across multiple evaluation metrics. Comprehensive analyses validated the effectiveness and advantages of all key modules in SG-Net, further demonstrating the potential and robustness of our approach in polyp segmentation.

CRedit authorship contribution statement

Qiaohong Chen: Conceptualization, Supervision, Writing – review & editing. **Zhenyang Xu:** Methodology, Writing – original draft, Software. **Xian Fang:** Funding acquisition, Writing – review & editing. **Qi Sun:** Visualization, Investigation. **Xin Wang:** Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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